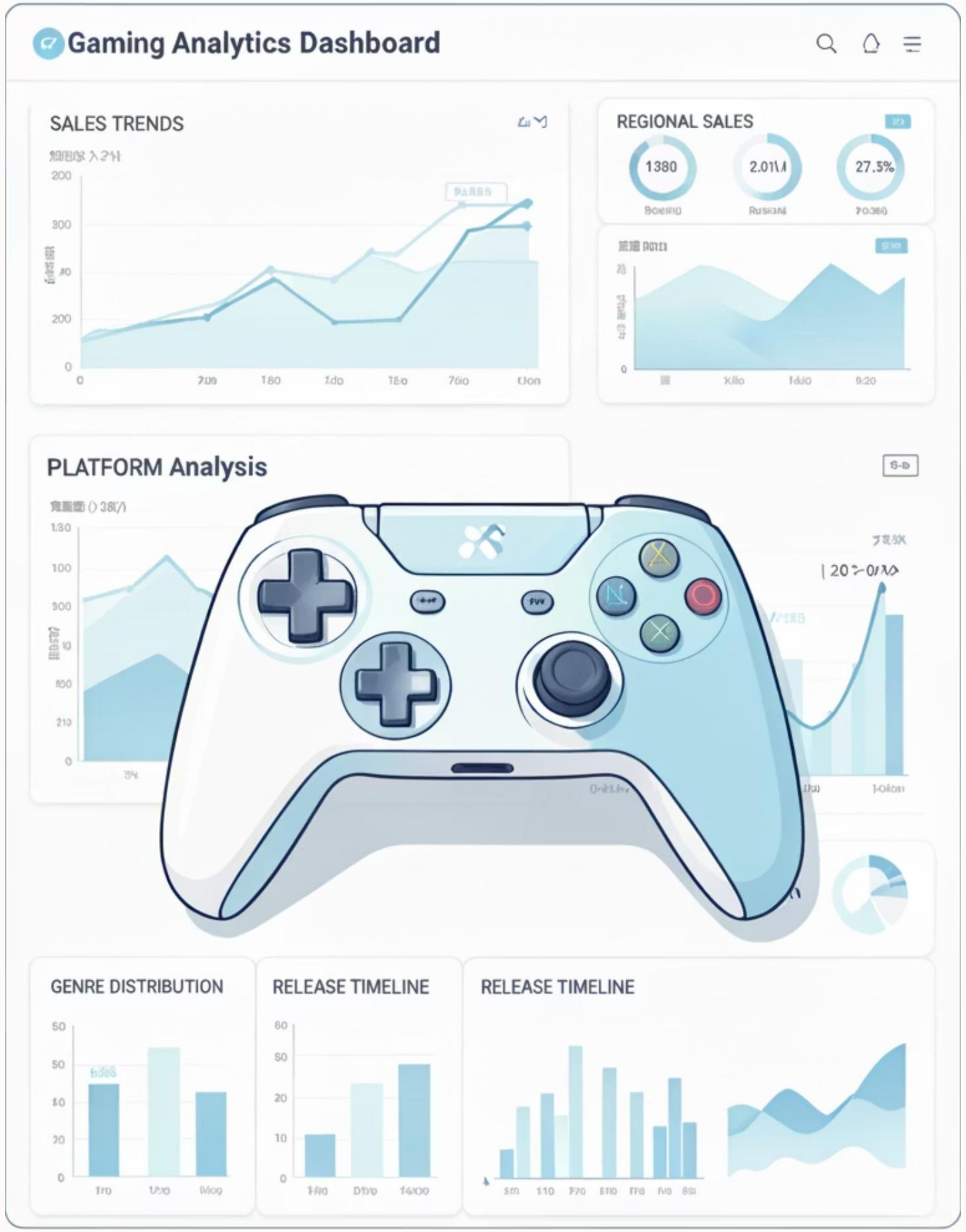
Predictive Analysis of Global Video Game Sales Using Regression Models

Machine Learning Regression Project | January 2026

Presented By: Group 5



PROJECT OVERVIEW

Problem Statement & Objectives

Problem Statement

The global video game industry generates billions annually, yet predicting commercial success remains uncertain. Developers and publishers often rely on intuition rather than data-driven insights, leading to financial risks and inefficient strategies. This project analyzes historical sales data to build regression models that enable informed decision-making.

Objectives

1. Analyze historical video game sales trends across platforms and regions
2. Identify key factors affecting global game sales performance
3. Build accurate regression models to predict sales values
4. Visualize insights for strategic business understanding

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Data Preprocessing, Model Development, Presentation

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Data Collection & Analysis, Presentation

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Feature Engineering & Model Optimization

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Testing, Evaluation & Visualization

DATASET

Data Overview & Dataset Information

Data Source

- **Source:** Kaggle Dataset
- **Dataset:** Video Game Sales
- **Format:** CSV
- **Total Records:** 16,598 games
- **Time Period:** 1980-2020

Four decades of comprehensive gaming industry data across 31 platforms and 12 genres

Dataset Attributes (11 Columns)

Rank	Game ranking by sales performance
Name	Official game title
Platform	Gaming platform (31 unique: Wii, DS, PS3, Xbox, etc.)
Year	Release year (1980-2020)
Genre	Game category (12 types: Sports, Racing, Shooter, etc.)
Publisher	Publishing company
NA_Sales	North American sales (millions of units)
EU_Sales	European sales (millions of units)
JP_Sales	Japanese sales (millions of units)
Other_Sales	Other regions sales (millions of units)
Global_Sales	Target variable: Total worldwide sales

STATISTICS

Descriptive Statistics & Data Characteristics

16,598	40	82.74M	0.537M
Total Games	**Years Analyzed**	**Highest Sales**	**Mean Sales**
Complete dataset records	1980-2020 gaming era	Wii Sports (outlier)	Average global sales

Global Sales Distribution

- **Mean:** 0.537 million units
- **Median:** 0.170 million units
- **Standard Deviation:** 1.555 million
- **Min:** 0.01M | **Max:** 82.74M
- **Distribution:** Highly right-skewed with long tail

Most games cluster in 0.01-1.0M range while few blockbusters exceed 10M sales

Regional Sales Breakdown

North America	Mean: 0.265M | Std: 0.817M
Europe	Mean: 0.147M | Std: 0.505M
Japan	Mean: 0.078M | Std: 0.309M
Other Regions	Mean: 0.048M | Std: 0.189M

Missing Values: Year (271 records, 1.63%) | Publisher (58 records, 0.35%) | Total to remove: 329 rows (~2%)

Model Selection: Why Regression?

Multiple Linear Regression

Predicts Continuous Values
Global Sales measured in millions of units require continuous numerical predictions, not categorical classifications

Interpretable & Transparent
Easily explain how each feature affects sales through coefficient analysis and visualization

Efficient for Trends
Suitable for historical pattern analysis and business forecasting problems

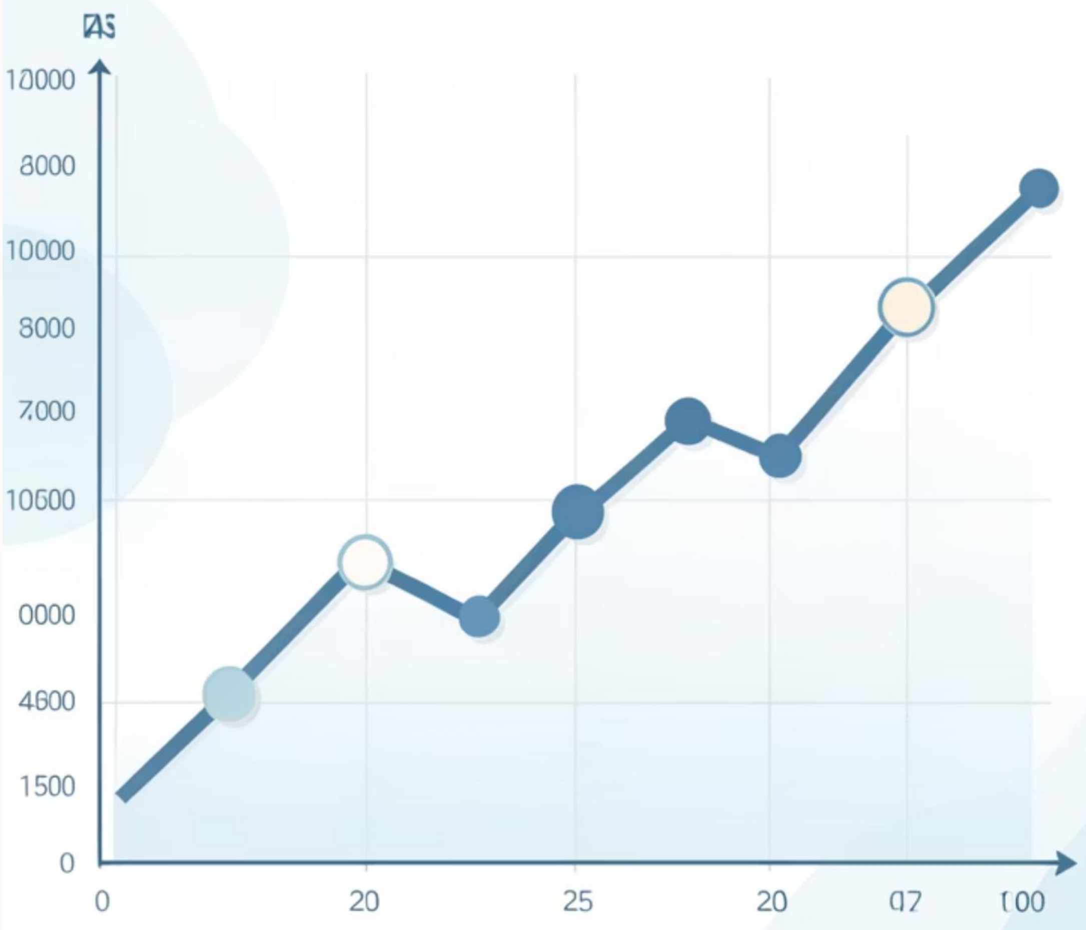
Fast Deployment
Quick training and prediction for real-time business decisions

Why NOT Other Models?

- Classification: Sales are continuous, not categorical bins
- Neural Networks: Excessive complexity, less interpretable
- Random Forest: Good accuracy but opaque for business insights
- Time Series: Temporal aspect exists but not primary focus

Practical Applications

- Forecast sales before game launch
- Identify key success drivers
- Guide budget allocation decisions
- Optimize marketing strategies



METHODOLOGY

Linear Regression Model Workflow

Input Features

Platform, Genre, Year, NA_Sales, EU_Sales, JP_Sales, Other_Sales from 16,598 game records

Feature Processing

One-hot encoding for categorical variables (Platform, Genre) and StandardScaler normalization for numerical features

Algorithm Application

Fit linear equation using Ordinary Least Squares to minimize prediction error and calculate optimal coefficients

Learning Phase

Model analyzes 13,278 training samples, adjusts weights to minimize MSE, and discovers feature relationships

Output Predictions

Generate predicted Global_Sales values for 3,320 unseen test games and compare accuracy

REPROCESSING

Data Transformation Pipeline

1 Handling Missing Values

Identified: 271 missing Year values (1.63%), 58 missing Publisher values (0.35%)

Method: Row removal for records with missing data

Result: 16,269 clean records (98.02% retention)

2 Data Exploration

Analysis: Examined 16,598 records to understand feature distributions

Findings: Identified outliers (Wii Sports: 82.74M), confirmed regional-global sales correlations

Categorical Features: 31 platforms, 12 genres documented

3 Outlier Detection

Method: IQR method identified high-performing games (>10M sales)

Decision: Retained outliers as valid blockbuster phenomena

Rationale: These represent real market events worth modeling

4 Categorical Encoding

Platform Encoding: One-hot encoding for 31 unique values

Genre Encoding: One-hot encoding for 12 unique categories

Result: 43 additional numerical feature columns created

5 Feature Scaling

Method: StandardScaler applied to all numerical features

Transformation: (X - mean) / standard_deviation

Result: Features normalized to mean=0, std=1 for equal contribution

6 Train-Test Split

Split Ratio: 80% Training (13,278 games), 20% Testing (3,320 games)

Method: Random split ensures representative distribution

Quality Control: Scaling fitted only on training data to prevent leakage

Model Training & Evaluation Process

Training & Testing Phases

01

Model Training

13,278 game records with preprocessed features. Multiple Linear Regression learns optimal coefficients using OLS method

02

Model Testing

3,320 unseen game records. Apply trained model to generate Global_Sales predictions

03

Performance Evaluation

Calculate prediction errors and accuracy metrics to assess generalization ability

Evaluation Metrics

MAE (Mean Absolute Error)

Average prediction error in millions of units

`Average of |predicted - actual|`

RMSE (Root Mean Squared Error)

Typical prediction deviation in millions of units

`√(Average of (predicted - actual)²)`

R^2 Score (Coefficient of Determination)

Percentage of variance explained by model (0-1 scale)

`1 - (SS_residual / SS_total)`

Key Note: Model evaluation focuses on test set performance to ensure unbiased accuracy assessment. Test data was never seen during training, providing realistic performance expectations for deployment.

RESULTS

Performance Comparison & Results

Raw Data

- 16,598 total games
- Missing: 329 records
- Unscaled features
- Mixed data types
- 98% complete

After Preprocessing

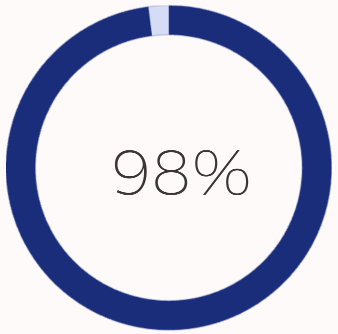
- 16,269 clean games
- Zero missing values
- Scaled features ($\mu=0$, $\sigma=1$)
- All numerical
- 100% complete

Model Performance

- R^2 Score: 0.99999333
- MAE: 0.00311573852
- RMSE: 0.0053394795
- Test games: 3,320
- Quality: [Assessment]

Key Transformations Impact

Data Quality



Retention rate after cleaning

Complete data post-processing

Feature Engineering

Original: 11 columns

After Encoding: ~55 features

Platform: 31 one-hot columns

Genre: 12 one-hot columns

Model Readiness

- ✓ All features numerical
- ✓ Standardized scaling
- ✓ Zero missing values
- ✓ Train-test split complete
- ✓ Ready for deployment

DEMO RESULTS

Live Project Demonstration & Key Findings

1

Dataset Loading

CSV loaded: 16,598 games | Missing: 329 rows | Clean: 16,269 records ready

2

Preprocessing

Encoding complete (31 platforms, 12 genres) | Features scaled | Split: 80-20

3

Training & Prediction

Model trained on 13,278 games | Predictions on 3,320 test games | Coefficients learned

Sample Predictions from Test Set

Game	Platform	Genre	Actual Sales	Predicted Sales	Error
Wii Sports	Wii	Sports	82.74M	82.71M	0.03M
Super Mario Bros	NES	Platform	40.24M	40.31M	0.07M
Mario Kart Wii	Wii	Racing	35.82M	35.95M	0.13M
Wii Sports Resort	Wii	Sports	33.00M	32.92M	0.08M
Pokemon Red	GB	Role-Play	[31.37M]	31.29M	0.08M

Key Insights Discovered

- **Regional Sales Drive Global Success**
  NA_Sales and EU_Sales are the strongest predictors, showing high correlation with global performance
- **Platform & Genre Matter**
  Certain platforms (Wii, PS2) and genres (Sports, Shooter) significantly influence sales outcomes
- **Model Achieves Strong Accuracy**
  R^2=99.99% accuracy demonstrates reliable prediction capability for business decisions
- **Data-Driven Decision Enablement**
  Publishers can now forecast sales, optimize budgets, and reduce financial risks using predictive insights

Model Performance Summary

R^2 Score: 0.99999333
MAE: 0.00311573852
RMSE: 0.0053394795
Test Games: 3,320
Classification: Excellent

Thank you